Vedant R. Patil

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Education

Veermata Jijabai Technological Institute (VJTI) Mumbai

Bachelor of Technology (B.Tech), Computer Engineering *2024 – 2028*

Technical Skills

- **Languages**: C++, Python, JavaScript, SQL

  Frameworks/Libraries: FastAPI, Streamlit, TensorFlow, Scikit-Learn, OpenCV, NumPy, Pandas, SciPy

  Databases: MariaDB, MySQL

  Tools: Git, GitHub, VS Code, Postman, Jupyter Notebook, Google Colab

  Technologies: Artificial Intelligence, Machine Learning, Deep Learning, REST APIs, Blockchain, Ethereum, IPFS

Projects

MedBlocks | ***FastAPI, MariaDB, Ethereum, IPFS*** 7 Months

- Developed the backend of a blockchain-enabled medical record management system using FastAPI for secure and scalable healthcare applications.
- Designed RESTful APIs for uploading, retrieving, and managing patient medical records with efficient backend architecture.
- Implemented Role-Based Access Control (RBAC) to enable secure access for patients and healthcare professionals.
- Integrated IPFS for decentralized file storage while maintaining structured metadata in MariaDB to ensure efficient record management.
- Collaborated with a multidisciplinary team to build a secure healthcare platform leveraging blockchain technology and decentralized storage.

SENTINEL-B | ***Python, Scikit-Learn, TensorFlow, Docker*** Ongoing

- Building the ML detection core of a non-IoC compromise detection system for firewalls, routers, and hosts for a national NTRO/NCIIPC cybersecurity hackathon.
- Implementing per-entity and peer-group behavioral baselining using Isolation Forest, ECOD, and sequence-based anomaly models to flag deviations without relying on signatures or IoCs.
- Designing ATT&CK-aligned invariant detectors, including beacon periodicity, DNS tunneling signal, and credential-access lineage analysis, to add attack-specificity beyond raw anomaly scores.
- Calibrating detector outputs into statistically fused risk scores using extreme value theory-based p-values to keep false alerts within an analyst-manageable budget.
- Collaborating with a multidisciplinary team to build and validate the system across host, network, and firewall/router telemetry.

CityPulse | ***MariaDB, SQL*** 6 Months

- Independently designed and implemented a normalized relational database for a community issue management platform.
- Developed a structured database schema consisting of multiple interconnected entities while ensuring normalization and referential integrity.
- Implemented SQL joins, views, triggers, stored procedures, and aggregation queries to automate workflows and improve query efficiency.
- Designed ER diagrams and database architecture supporting scalable complaint registration, issue tracking, and reporting.
- Optimized relational database operations through effective schema design and indexing strategies.

Relevant Coursework

- Data Structures & Algorithms • Database Management Systems • Object-Oriented Programming • Operating Systems • Computer Networks • Software Engineering

Certifications

- AICTE Lenovo LEAP NextGen Scholar — Generative AI & Agentic Systems Engineering